Heterogeneous Expert PLE: An Explainable Multi-Task Architecture for Financial Product Recommendation

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Abstract. Financial product recommendation requires not only predictive accuracy but also regulatory-compliant explainability. Existing multi-task learning (MTL) approaches (MMoE, PLE) are validated on 2–4 homogeneous tasks, leaving open whether they scale to production-scale heterogeneous workloads where binary, multiclass, and regression tasks compete for shared capacity. We present three contributions. **First**, we propose *Heterogeneous Expert PLE*, where seven architecturally distinct experts (DeepFM, Temporal Ensemble, Hyperbolic GCN, PersLay, Causal, LightGCN, Optimal Transport) share a common basket with a FeatureRouter assigning each expert its designated feature groups, providing a structural guarantee against expert collapse and inherent explainability through business-interpretable gate weights. **Second**, through ablation on a 13-task benchmark (7 binary + 3 multiclass + 3 regression, 1M customers), we discover that *loss-level* inter-task transfer (adaTT) degrades performance in this heterogeneous setting due to 156 task-pair affinity estimation instability, while the single largest improvement comes from correcting a subtle uncertainty-weighting implementation gap where per-task loss weights were silently ignored. **Third**, we experiment with *GradSurgery*, a task-type gradient projection method that replaces loss-level transfer with gradient-level conflict resolution, reducing the 156 task-pair problem to 3 task-type-group projections. However, GradSurgery shows no meaningful advantage over the PLE-only baseline while requiring significantly more VRAM (due to the retained computation graph), and is not adopted for production. Ablation results: PLE with softmax gating achieves the best NDCG@3 (0.714) among gate variants, softmax outperforms sigmoid in the heterogeneous setting (reversing findings from homogeneous-task literature), and GradSurgery maintains baseline performance (AUC 0.673, F1-macro 0.203) compared to adaTT's degradation of −0.019 AUC. The operational motivation — consolidating 13 individual models into a single MTL model for unified training, serving, and monitoring — is validated: the shared-bottom baseline already exceeds per-task XGBoost ceilings, and PLE provides additional gains on ranking metrics without sacrificing classification performance.

Keywords: Multi-task learning, Progressive Layered Extraction, Mixture of Experts, Financial recommendation, Explainable AI, Ablation study

1 Introduction

1.1 Problem Definition

Financial product recommendation differs fundamentally from e-commerce or content recommendation. The primary deliverable is not a probability score but a *reason that the customer can accept*. Three audiences must be persuaded: customers ("Why this product for me?" — trust leads to conversion), relationship managers ("Why recommend this to this customer?" — sales justification), and regulators [7, 9] ("Why was this decision made?" — compliance obligation). Korea's AI Basic Act [27] further classifies financial recommendation as potentially high-impact AI.

Existing approaches fall short on this persuasion requirement. Single-task models cannot jointly predict churn, product affinity, and customer lifetime value [3]. Model ensembles (N separate models) multiply management overhead and serving cost, and "MLP #3 contributed 28%" provides no business meaning. Post-hoc explanations (SHAP, LIME) are decoupled from model internals, computationally expensive at serving time, and demonstrably unstable under input perturbation [22, 34, 35].

1.2 Core Insight

Conventional recommendation models function as "black-box shakers" — features go in, probabilities come out, with only statistical correlations as justification. As Pearl [32] argues, there is a fundamental gap between *seeing* (association) and *understanding* (causation). Humans are not persuaded by correlations; they require *causal narratives* [31].

If each expert in a multi-task model captures a *different kind of "why"* — temporal trends, hierarchical product structure, causal pathways, collaborative patterns — then the gating mechanism itself becomes an explanation: "This recommendation is driven primarily by your

spending trend (Temporal, 35%) and product category fit (HGCN, 28%)."

This is the founding design principle of our architecture.

The key realization is that financial recommendation is not about a single prediction ("will this customer buy?") but about **understanding a customer as a whole person** from multiple perspectives simultaneously: Will they churn? (lifecycle). How much will they spend? (value). What product fits them? (consumption). How do they behave? (engagement). Each question demands a different analytical lens — temporal analysis cannot answer hierarchical product fit, and graph-based collaborative signals cannot predict spending trajectories. This is why heterogeneous experts are not merely a performance optimization but a structural necessity for multi-faceted customer understanding.

This perspective aligns with a broader shift in both academia and regulation. Pearl's *Ladder of Causation* [31] distinguishes three levels: association, intervention, and counterfactuals. Most recommendation systems operate at level 1 (association); our architecture operates primarily at level 1 (association), with the Causal expert approaching level 2 (intervention) for tasks where its gate weight is dominant. Regulatory frameworks — the EU AI Act, Korea's FSS guidelines and AI Basic Act — increasingly demand this shift toward structurally transparent explanations (detailed regulatory analysis in Section 2.4).

1.3 Contributions

1. **Heterogeneous Shared Expert Basket with Structural Collapse Guarantee**: We replace PLE's homogeneous MLP experts with seven architecturally distinct experts (DeepFM, Mamba+LNN+Transformer, HGCN, PersLay, NOTEARS, LightGCN, Optimal Transport). Unlike prior "heterogeneous" MoE work that varies expert *size* [41] or *modality* [1], we vary the fundamental *inductive bias*, providing a structural guarantee against expert collapse — a persistent failure mode in homogeneous MoE/PLE deployments [19].

2. **FeatureRouter: Heterogeneous Architecture × Heterogeneous Input**: Beyond architectural diversity, each expert receives only its designated feature groups (declared via expert routing declarations at the feature group level), not the full 349D input. This "heterogeneous architecture × heterogeneous input" design eliminates irrelevant features per expert (per-expert dims: 27D–168D), reducing model parameters from 4.77M to 2.8M while strengthening each expert's specialization.

3. **Inherent Explainability**: Because each expert encodes a named mathematical operation (not a generic MLP), CGC gate weights directly yield business-interpretable explanations without post-hoc attribution methods.

4. **Multi-disciplinary Feature Engineering**: Features derived from eleven academic disciplines — including unconventional applications of chemical kinetics (spending activation rate), epidemic modeling (product adoption diffusion), criminological Routine Activity Theory (transaction regularity), and wave interference (spending periodicity) — serve dual roles as learning signals and recommendation context that is reverse-mapped to business language for customer-facing explanations.

5. **Financial DNA Task Grouping**: Four task groups (engagement, lifecycle, value, consumption) aligned with the two-axis decomposition (Financial DNA × Data Modality), providing semantic routing for expert selection.

6. **Gate Type Analysis for Heterogeneous MTL**: We demonstrate that softmax gating outperforms sigmoid in 13-task heterogeneous settings (7 binary + 3 multiclass + 3 regression), reversing the conventional preference from homogeneous-task literature [28, 39]. The reversal is attributed to softmax's protective isolation of minority-type tasks from majority-type gradient corruption.

7. **Loss-Level vs. Gradient-Level Transfer**: We show that adaTT (loss-level transfer) degrades performance at 13-task scale due to 156 task-pair affinity estimation instability, and experiment with GradSurgery (gradient-level projection between 3 task-type groups) as an alternative that avoids adaTT's degradation but shows no additional gain and incurs VRAM overhead from the retained computation graph; GradSurgery is not adopted for production.

8. **Uncertainty Weighting Correction**: We identify and fix a subtle implementation gap where per-task loss weights were silently ignored under uncertainty weighting — yielding the single largest performance improvement (+0.018 NDCG@3, +0.031 F1-macro), larger than any architectural change.

9. **Comprehensive Ablation**: 27 scenarios (18 joint feature+expert + 9 structure cross) on a reproducible 1M-customer benchmark with Gaussian Copula + latent variable variance budget.

10. **Config-driven Pipeline**: End-to-end system (feature engineering → training → distillation → serv-

ing) controlled by two configuration files, enabling deployment by teams with 1–2 ML engineers.

11. **Reproducible Benchmark**: Synthetic data generation with variance budget [30] for controlled AUC ceilings, validated against XGBoost baselines, with open-source code and fixed seeds.

2 Related Work

2.1 Multi-Task Learning for Recommendation

The progression from Shared-Bottom [3] to MMoE [23] to PLE [39] reflects increasing sophistication in managing negative transfer between tasks. MMoE introduced per-task gating over a shared expert pool, while PLE further separated shared and task-specific experts with a progressive extraction structure. AdaTT [20] extended this with adaptive inter-task transfer.

Other notable MTL architectures include ESMM [24] (entire-space modeling for conversion rate), STAR [37] (star topology for multi-domain CTR), and M3oE [42] (multi-domain multi-task MoE with AutoML structure search).

However, all prior MoE/PLE architectures employ **homogeneous experts** — multiple MLPs with identical architecture but different initializations. This limits the diversity of learned representations to what parameter variation alone can achieve, and leads to a well-documented failure mode: **expert collapse**.

2.2 Mixture of Experts and the Expert Collapse Problem

The MoE paradigm [36] and its successors (Switch Transformer [8]) demonstrate the power of conditional computation. However, homogeneous MoE architectures suffer from **expert collapse**, where all gates converge to routing inputs to a single expert, effectively reducing the MoE to a shared-bottom architecture. Pinterest Engineering reported that their MMoE deployment "would oftentimes collapse into a state where all tasks use the same single expert." Kuaishou's HoME [19] confirmed that "expert collapse occurs when all gates assigned larger weights to a single shared expert and almost ignored other shared experts."

Existing mitigations operate post-hoc: Gram-Schmidt orthogonalization to force diverse expert representations, expert normalization to align output distributions, and load-balancing losses to distribute routing. These are engineering patches on a structural problem — homogeneous experts have no inherent reason to specialize differently because their only source of diversity is random initialization.

Recent work has explored limited forms of heterogeneity: Jamba [1] combines Transformer and Mamba layers within MoE for LLMs (2 architecture types), and MOWST [41] pairs lightweight MLP "weak experts" with GNN "strong experts" in a hierarchical arrangement (2 types, unequal roles). M3oE [42] separates experts by domain but uses identical MLP architectures within each domain. MoE++ introduces computation-level heterogeneity (zero-compute vs. FFN experts) but not architectural heterogeneity.

Furthermore, recent theoretical work demonstrates that the standard softmax gating mechanism itself contributes to representation collapse. Nguyen et al. [28] prove that sigmoid gating achieves higher sample efficiency than softmax gating in MoE models: softmax's sum-to-one constraint forces experts into unnecessary competition, where increasing one expert's weight requires decreasing another's, leading to representation collapse even when experts provide complementary signals. Sigmoid gating eliminates this inter-expert competition by allowing each expert's contribution to be evaluated independently, achieving faster convergence and better utilization of all experts. This finding is especially relevant when experts are *heterogeneous*: since their outputs have fundamentally different scales and distributions (e.g., hyperbolic embeddings vs. persistence diagrams), softmax competition is particularly harmful — suppressing a topological expert to increase a temporal expert's weight means losing irreplaceable topological signal, not merely a redundant representation.

Gap: No prior work composes experts with **fundamentally different inductive biases** — graph convolution, state-space models, topological persistence, causal DAG constraints, optimal transport — as equal peers within a shared expert basket. Expert collapse manifests in two forms: (1) *function space collapse* — homogeneous experts converge to identical functions due to shared architecture and gradient dynamics; (2) *routing collapse* — gate weights concentrate on a single expert, effectively deactivating others regardless of their functional diversity. Our heterogeneous expert design structurally prevents (1). For (2), prior work suggests sigmoid gating [28] mitigates routing collapse through independent expert contributions. However, our ablation (Section 5.3) reveals that in heterogeneous task settings, softmax's competitive isolation better protects minority-type tasks — a reversal of the homogeneous-task finding.

Our work provides a **structural guarantee** against expert collapse: a DeepFM expert cannot converge to the

same function as an HGCN expert regardless of training dynamics, because their architectures encode fundamentally different mathematical operations.

2.3 Explainability in Recommendation

SHAP [22] and LIME [34] provide model-agnostic explanations but suffer from instability, computational cost, and disconnect from model internals. Integrated Gradients [38] offers theoretically grounded attribution but still operates post-hoc.

Gap: Structural (inherent) explainability that produces business-meaningful explanations as a natural byproduct of the forward pass, without additional computation.

2.4 Financial Recommendation and Regulation

Recent work on deep learning for financial product recommendation [5, 26] demonstrates improvements over collaborative filtering, while Martinez-Plumed et al. [25] show that sequential models on longitudinal transaction data improve cross-selling prediction. However, none of these systems address regulatory explainability requirements.

The EU AI Act [7] classifies financial recommendation as high-risk AI, mandating transparency (Art. 13), human oversight (Art. 14), and robustness (Art. 15). Korea's FSS guidelines [9] and AI Basic Act [27] impose similar requirements. Current approaches rely on post-hoc SHAP/LIME for regulatory compliance, which has documented limitations in financial contexts [35].

Gap: No recommendation system provides verifiable mapping from regulatory requirements to architecture components, with explanations generated structurally rather than post-hoc.

3 Architecture

3.1 Reductionist Framework: Two Axes of Decomposition

Before describing the architecture, we present the analytical framework that governs all subsequent design decisions.

The complexity of financial customer understanding is decomposed along two orthogonal axes:

Axis 1: Who is the customer? (Financial DNA)

We decompose the question "who is this customer?" into four irreducible dimensions, each capturing a fundamentally different aspect of customer identity:

[1]

nba_primary target design. `nba_primary` targets *product groups* (7 classes), not individual product indices. The 7 classes are: 0 = no NBA, 1 = savings/guarantee, 2 = checking accounts, 3 = deposits, 4 = investments, 5 = credit loans, 6 = debits. This redesign (from 25 individual-product classes) achieves better class balance and enables meaningful ranking-based evaluation. The original 25-way classification had a long-tail distribution in which rare products dominated macro-F1, making standard F1-macro unreliable; NDCG@K is appropriate for the group-level ranking objective.

Axis 2: What form does the information take? (Data Modality)

Customer data exists in structurally distinct modalities, each requiring a different mathematical tool to extract meaningful signals:

The cross-product of these two axes defines the architecture: each (DNA × Modality) cell determines which expert processes which features for which task

DNA	Question	Tasks
Engagement	What do they *do*?	next_mcc mcc_diversity_trend top_mcc_shift
Lifecycle	Where *are* they?	churn_signal segment_prediction
Value	How much are they *worth*?	cross_sell_count product_stability
Consumption	What *will* they buy?	will_acquire_* (5) nba_primary

Table 1: Financial DNA decomposition (Axis 1). Four irreducible dimensions of customer identity.

[1]Four tasks present in early development (income tier, tenure stage, spend level, engagement score) were removed after identifying them as deterministic feature transformations that do not constitute genuine prediction objectives.

Modality	Examples	Expert	Generator
State	is_active, holdings	DeepFM	base
Snapshot	balance, demographics	GMM, DeepFM	GMM clustering
Short-term series	recent transactions	Transformer	temporal
Long-term series	monthly trends	Mamba	mamba temporal
Disrupted series	dormant→active	LNN	model derived
Hierarchy	MCC merchant category tree	HGCN	merchant_hierarchy
Relations	customer-product graph	LightGCN	graph collab.
Topology	behavioral shape	PersLay	TDA global/local
Causality	behavioral causation	Causal	causal features

Table 2: Data modality decomposition (Axis 2). Each modality is mapped to an expert and feature generator.

group. The 7 heterogeneous experts, 12 feature groups, and 4 task groups are not arbitrary design choices but necessary consequences of this two-axis decomposition. A homogeneous MLP expert basket ignores Axis 2 entirely — it treats hierarchical, temporal, and topological data identically — which is why it cannot achieve the task-type-specific specialization that our ablation demonstrates.

3.2 Design Philosophy

The architecture emerged from severe real-world constraints at a Korean public financial institution. A team of three — one data scientist serving as PM, with GARP Financial Risk Manager (FRM) certification and a career spanning credit/market risk analysis, regulatory compliance, MyData licensing, big data platform construction and operations, data science projects, and recommendation system management, plus two engineers — needed to replace a legacy ALS-based collaborative filtering system with a next-generation recommendation model.

The constraints were formidable: no dedicated ML infrastructure budget, a single desktop-grade GPU (NVIDIA RTX 4070, 12GB VRAM) as the only training hardware, no GPU inference servers for deployment, and strict regulatory requirements (Korean FSS AI guidelines, EU AI Act).

Rather than treating these constraints as limitations, the team adapted its methodology at every level:

- **AI-augmented development** using Claude (Anthropic), Gemini (Google), and Cursor, with each team member leading a parallel team of AI agents;
- **Parameter-efficient architecture design** where structural inductive biases replace the brute-force capacity of large MLPs;
- **Knowledge distillation** to LGBM for GPU-free CPU inference on AWS Lambda;
- **Config-driven pipeline** requiring only two YAML files to control the entire system, enabling operation by a minimal team.

Early exploration considered a Black-Litterman-inspired approach, treating multiple models' predictions as "expert views" combined via Bayesian updating. This was abandoned because the Bayesian combination process obscured each model's contribution, making business-meaningful explanation impossible — a critical failure for regulatory compliance.

The key reframing was: instead of combining *models* externally, combine *experts* internally within a single model. This led to the selection of PLE [39] with a critical modification: replacing homogeneous MLP experts with structurally heterogeneous experts, each encoding a different inductive bias.

3.2.1 Structural Isomorphism: Why Cross-Disciplinary Methods Work

The multi-disciplinary feature engineering in this work is not arbitrary borrowing from unrelated fields. It is grounded in *structural isomorphism* — the observation that mathematically equivalent structures recur across different domains, enabling tools developed in one field to be rigorously applied to structurally equivalent problems in another.

Consider: a dormant customer resuming spending follows the same dynamics as a chemical reaction overcoming an activation energy barrier. Product adoption spreading through a customer network follows the same compartmental dynamics as an SIR epidemic model. These are not metaphors — the governing equations are identical, and the solutions transfer with mathematical rigor. We emphasize that these are not loose analogies: the governing differential equations are structurally identical (see Appendix E for side-by-side comparison). The validity of

each mapping can be independently verified by checking whether the mathematical preconditions of the source domain's theorems hold in the financial domain.

This principle has a distinguished history in science and engineering: Shannon's information entropy was directly borrowed from Boltzmann's thermodynamic entropy; the Black-Scholes option pricing model is derived from the heat diffusion equation; Google's PageRank is an application of Markov chain stationary distributions. In each case, recognizing structural isomorphism between two domains enabled a breakthrough that neither domain could have achieved alone.

Our contribution is applying this principle *systematically* to financial customer understanding: for each aspect of customer behavior, we identify a domain whose mathematical tools are structurally isomorphic to the underlying dynamics, and instantiate that domain's methods as both features (for prediction) and expert architectures (for interpretation). The multi-disciplinary approach is not a collection of ad-hoc features but a principled application of analogical reasoning through structural equivalence.

Four principles guide the architecture:

1. **Robust Explainability**: Model structure itself generates explanations. Each expert's name carries business meaning (e.g., "Temporal" = time patterns, "HGCN" = product hierarchy), so CGC gate weights are directly interpretable without post-hoc methods.
2. **Expert Contribution Diagnostics**: The heterogeneous design enables per-expert contribution analysis — identifying which inductive biases are beneficial vs. harmful in a given data regime. Homogeneous experts would show uniform, uninformative degradation.
3. **Flexible Extensibility**: New features, tasks, or experts are added via YAML configuration, not code changes. The system has been extended from 4 to 13 tasks without architectural modification.
4. **Unified Manageability**: The entire pipeline (feature engineering → training → distillation → serving → monitoring) is controlled by two configuration files, reducing operational overhead for teams with limited ML engineering resources.

3.3 Data Axis Classification

Financial customer data exhibits inherently multi-modal structure. We classify data along multiple axes, each mapped to an optimal feature generator and expert:

The complete data axis to expert to feature generator mapping is shown in Table 2.

Note: Short-term, Long-term, and Disrupted series map to the three sub-components of the Temporal Ensemble expert (Transformer, Mamba, LNN respectively). GMM is a feature generator, not a standalone expert.

3.4 Heterogeneous Expert Basket

Unlike standard PLE where shared experts are identical MLPs, our shared expert basket contains seven struc-

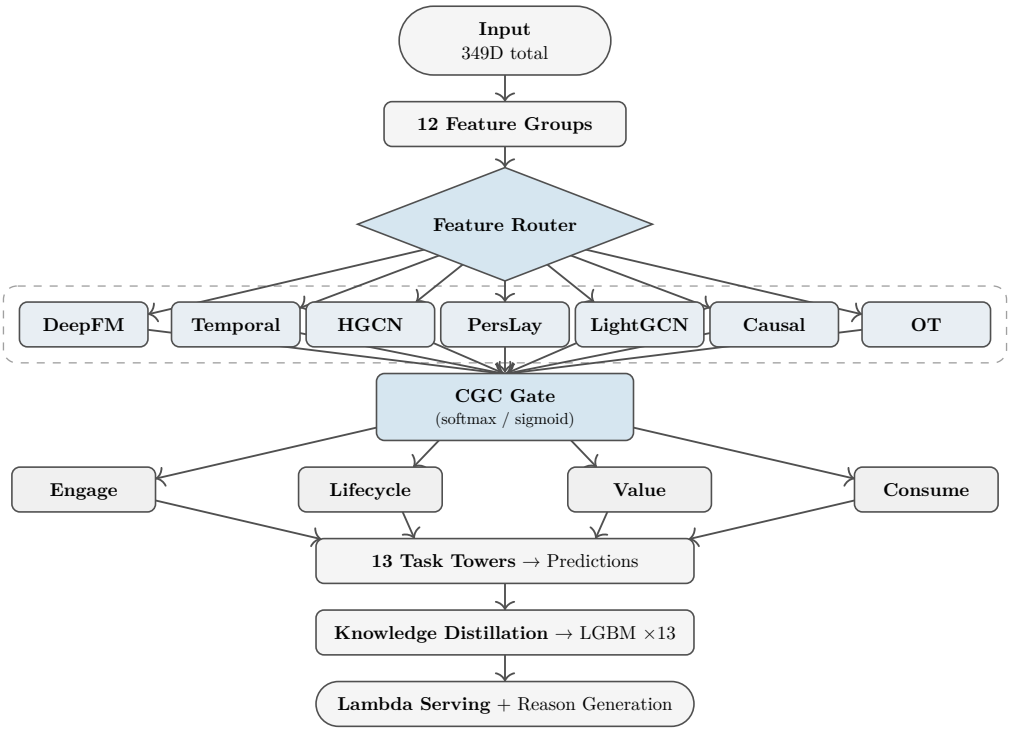

Figure 1: Heterogeneous Expert PLE architecture overview.

Expert	Inductive Bias	Captures
DeepFM [12]	Feature interaction	2nd-order cross features
Temporal Ensemble	Multi-scale temporal	Short/long/disrupted series
HGCN [4]	Hyperbolic hierarchy	MCC merchant category tree (Poincaré)
PersLay [2]	Topological persistence	Behavioral shape patterns
LightGCN [14]	Graph convolution	Collaborative filtering
Causal [43]	DAG constraint	Causal direction between features
Optimal Transport [6]	Distribution matching	Segment distribution shifts

Table 3: Seven heterogeneous experts with distinct inductive biases.

turally distinct networks. Each expert corresponds to one or more modality axes (Table 2), ensuring that every structurally distinct data type is processed by an architecture designed for it:

Because each expert has a named inductive bias, the gate weights tell the customer *why* in business terms — not "hidden unit 47 activated" but "your spending trend drove this recommendation." Each expert was selected based on a specific gap in financial customer understanding that no other expert type addresses:

- **DeepFM** [12]: Financial behavior is driven by feature interactions (e.g., income × product holdings × channel preference). FM's low-rank factorization computes 2nd-order crosses in $O(nk)$ vs. $O(n^2)$ for brute-force enumeration, while the Deep component captures higher-order interactions.
- **HGCN** [4]: The MCC merchant category hierarchy (10 L1 categories → 30 L2 subcategories → 109 leaf codes) is inherently tree-structured. Hyperbolic space (Poincaré ball, 8D) embeds trees with exponentially less distortion than Euclidean space [29] — critical for a 550K-node merchant hierarchy. HGCN receives `merchant_hierarchy` features (27D Poincaré embeddings) via FeatureRouter, not product co-holding features (those belong to LightGCN's `product_hierarchy` group).

  HGCN and LightGCN serve distinct roles that were inadvertently conflated in early experiments. HGCN embeds the MCC merchant category hierarchy (L1: 10 categories → L2: 30 subcategories) into Poincaré disk space, capturing tree-structured category relationships. LightGCN learns customer-product bipartite affinity via collaborative filtering on the 24-product co-holding graph. The `feature_groups.yaml` `target_experts` declarations enforce this separation: `merchant_hierarchy` → HGCN only, `product_hierarchy` → LightGCN only.
- **PersLay** [2]: Topological Data Analysis captures *shape* features (connected components H_0, cycles H_1, voids H_2) of customer spending patterns. H_1 loops reveal consumption cycles; H_2 voids detect systematic spending avoidance. These features are provably stable under noise (Stability Theorem).
- **Temporal Ensemble**: Financial time series combine regular snapshots, bursty transactions, and multi-month dormancy. Mamba [11] handles long-range trends in $O(n)$; LNN [13] adapts its time constants to irregular intervals; Transformer captures short-range attention patterns.
- **Causal** [43]: A learnable adjacency matrix $W \in \mathbb{R}^{d \times d}$ (where $d = 32$ compressed feature dimensions) encodes causal direction between features. During training, the NOTEARS acyclicity constraint ($\mathrm{tr}(e^{W \circ W}) - d = 0$, Taylor-approximated to 10th order) is added as a regularization loss, guiding W toward a DAG structure. At inference time, the learned W is frozen and applied as $\hat{z} = z + z \cdot (W \circ W)$, injecting discovered causal structure into the representation without rerunning the optimization. This enables "A causes B" explanations rather than "A correlates with B" [32].
- **LightGCN** [14]: Collaborative filtering via neighborhood aggregation on the customer-product bipartite graph. Stripped to essentials (no feature transform, no activation), which outperforms more complex GCN variants for recommendation.
- **Optimal Transport** [6]: Sinkhorn-regularized Wasserstein distance measures distributional shift between customer spending profiles and segment prototypes, respecting the metric structure of the feature space (unlike KL divergence).

The rationale for heterogeneous experts is also rooted in a hardware constraint: with a single desktop GPU (12GB VRAM), we cannot scale a homogeneous MLP expert to sufficient width/depth for high expressiveness.

Instead, each expert leverages a *structural inductive bias* to capture patterns that would require orders of magnitude more MLP parameters. For example, HGCN embeds the MCC merchant category hierarchy in 27 hyperbolic dimensions — achieving in $O(d)$ parameters what Euclidean embeddings require $O(2^d)$ dimensions to represent without distortion [4].

The total parameter count across all seven experts is **2.8M**, reduced from the 4.77M baseline by replacing broad shared inputs with expert-specific subsets via FeatureRouter. The diversity of learned representations is fundamentally richer despite the smaller parameter budget.

3.4.1 FeatureRouter: Heterogeneous Input × Heterogeneous Architecture

FeatureRouter is now active, implementing a stronger form of expert specialization: not only does each expert use a **different architecture** (heterogeneous basket), but each also receives a **different subset of input features** (heterogeneous input). This "heterogeneous architecture × heterogeneous input" design maximizes the signal-to-noise ratio for each expert by eliminating features that carry no information relevant to that expert's inductive bias.

Feature group assignments are declared in `feature_groups.yaml` via the `target_experts` field, ensuring zero dataset-specific hardcoding in model code. The resulting per-expert input dimensions are:

The sum of per-expert dimensions (754D) exceeds 349D because several feature groups are shared across multiple experts where complementary inductive biases benefit from the same signal (e.g., state features are useful to both DeepFM for interaction modeling and Causal for DAG structure inference).

3.4.2 CGC Gate Design: Softmax vs. Sigmoid

Standard PLE uses softmax CGC gates, which impose a competitive constraint: gate weights sum to 1, forcing experts to compete for influence. With homogeneous MLP experts this is reasonable — experts provide redundant representations and the gate selects the best one.

Expert	**Input Dim**	**Feature Groups Routed**
DeepFM	168D	• demographics • products • txn_behavior • derived_temporal • gmm • model_derived
Temporal Ensemble	139D	• txn_behavior • hmm • mamba • model_derived
HGCN	27D	merchant_hierarchy (MCC Poincaré embeddings)
PersLay	32D	tda_global, tda_local
Causal	161D	• demographics • products • txn • derived_temporal • product_hierarchy • gmm
LightGCN	100D	product_hierarchy, graph_collaborative
Optimal Transport	127D	• demographics • products • txn • derived_temporal • gmm

Table 4: Per-expert input dimensions after FeatureRouter. Total feature space: 349D (Phase 0 v3/v4).

With heterogeneous experts, this competition is harmful: each expert provides unique, non-redundant information (temporal $\neq$ hierarchical $\neq$ topological), and suppressing one means losing irreplaceable signal. Recent theoretical work [28] proves that sigmoid gating achieves higher sample efficiency by eliminating inter-expert competition.

We implement both gate types for empirical comparison:

- **Softmax**: $w_k = \text{softmax}(W \cdot h)_k$ — competitive, sum-to-one.
- **Sigmoid**: $w_k = \sigma(\boldsymbol{w}_k \cdot h) / \sum_j \sigma(\boldsymbol{w}_j \cdot h)$ — independent evaluation, normalized.

Note: Our sigmoid variant includes post-hoc normalization for comparability with softmax. Nguyen et al.'s theoretical analysis applies to unnormalized sigmoid; the normalized variant used here partially reintroduces competition through the denominator.

The sigmoid variant evaluates each expert's relevance independently before normalization, allowing multiple experts to receive high weight simultaneously. The theoretical expectation is that sigmoid benefits heterogeneous experts by preserving non-redundant signals. However, our structure ablation (Section 5.3) reveals a surprising reversal: softmax **outperforms** sigmoid in the 13-task heterogeneous setting, because its competitive allocation **protects** minority-type tasks (multiclass, regression) from gradient corruption by the majority type (binary). This finding challenges the theoretical prediction and highlights that task-type heterogeneity, not just expert heterogeneity, determines the optimal gate design.

3.4.3 Temporal Ensemble: Expert-within-Expert

Financial time series exhibit a uniquely complex structure rarely seen in other domains. A single customer may simultaneously have: monthly balance snapshots (regular, long-range), daily transaction sequences (irregular, bursty), and multi-month dormancy gaps (disrupted). No single temporal model handles all three well:

- **Mamba** [11] (State Space Model): Captures long-range dependencies with $O(n)$ efficiency via selective state spaces. Ideal for monthly/quarterly trends spanning 12+ months.
- **LNN** [13] (Liquid Neural Network): Adaptive time constants that naturally handle irregular sampling intervals and dormancy gaps. When a customer is inactive for 3 months then suddenly active, LNN's continuous-time dynamics adapt without requiring imputation.
- **Transformer**: Attention-based short-range context extraction. Captures patterns within the most recent 30–90 days of transaction sequences where positional relationships matter.

The three models' outputs are concatenated and projected, mirroring the heterogeneous expert philosophy at a finer granularity. This design is validated in the temporal component ablation, where removing any single temporal component degrades performance on different task groups.

3.5 Multi-disciplinary Feature Engineering

A distinguishing aspect of this work is the systematic application of methodologies from diverse academic disciplines to financial customer behavior. Now that the expert basket has been introduced, the rationale becomes clear: each discipline's features are designed to feed a *specific expert* whose inductive bias can best exploit that signal type. Rather than relying solely on standard statistical features (mean, variance, trend), we apply domain-specific mathematical tools that each extract a structurally different signal from the same underlying data.

Note: Phase 0 v3/v4 produces 349 total features (up from 316 in earlier versions). FeatureRouter routes feature subsets to each expert (per-expert dims from Phase 0 v3: DeepFM 168D, Temporal 139D, HGCN 27D, PersLay 32D, Causal 161D, LightGCN 100D, OT 127D); model parameters: 2.8M. Expert routing is built from `feature_groups.yaml` `target_experts` declarations (group-level), ensuring each expert receives only its designated feature groups.

Several of these applications are, to our knowledge, novel in financial recommendation:

Chemical kinetics models spending behavior as a reaction system: the "activation energy" represents the threshold for a dormant customer to resume spending, while "catalysts" (e.g., payroll deposits) lower this threshold. The half-life of spending intensity after a peak captures decay dynamics that simple moving averages cannot express.

SIR epidemic model treats product adoption as a contagion process: customers are "susceptible" (not yet adopted), "infected" (recently adopted), or "recovered" (churned). The basic reproduction number R_0 of a product category across a customer's social graph predicts adoption velocity.

Friedman's Permanent Income Hypothesis [10] decomposes observed income into permanent (stable, long-term) and transitory (bonus, irregular) components using HP filter or Kalman filter. This distinction is critical for financial recommendation: a customer with high transitory income should not be recommended long-term investment products that assume stable cash flow. The

Discipline	Method	Dim	Financial Interpretation
Topology	Persistent Homology (TDA)	32	Behavioral shape persistence: which consumption patterns are transient vs. structural
Hyperbolic Geometry	HGCN embedding	27	MCC merchant category distance: preserving hierarchy in low dimensions
Control Theory	Mamba (State Space)	50	Long-range behavioral dependencies: how past habits influence present
Stochastic Processes	HMM state transitions	25	Latent lifecycle stages: dormant → growing → mature → at-risk
Chemical Kinetics	Reaction rate modeling	6	Spending activation rate, half-life, dormancy reactivation catalysis
Epidemiology	SIR compartment model	5	Product adoption as "infection": susceptible → adopted → churned
Criminology	Routine Activity Theory	5	Transaction regularity: burstiness, circadian variance, routine breakpoints
Signal Processing	FFT + Hilbert transform	8	Spending periodicity: spectral entropy, harmonic power, phase locking
Economics	Friedman Permanent Income	8	Income decomposition: permanent vs. transitory income, consumption smoothing
Graph Theory	LightGCN	66	Collaborative filtering: similar customer behavioral transfer
Statistics	GMM soft clustering	22	Probabilistic segmentation: multi-modal customer distribution

Table 5: Multi-disciplinary feature engineering across 11 academic disciplines.

FD-TVS scoring system (detailed in companion paper) uses this decomposition to weight recommendations by income stability type.

These features serve a dual purpose beyond predictive contribution: they enrich the *recommendation context* that no standard feature can provide. Internally, "activation energy decrease" (chemical kinetics) signals rising re-engagement likelihood, and "growing SIR infection ratio" (epidemiology) signals accelerating product adoption — but the customer never sees these scientific terms. An `interpretation_registry` maps each scientific feature to business-interpretable language, and the LLM agent generates natural-language reasons accordingly (e.g., "Your recent activity pattern shows signs of recovery"). The scientific frameworks ground the model's internal reasoning; the customer receives business language.

3.6 Financial DNA Task Grouping

We organize 13 prediction tasks into four groups based on financial customer DNA. Each task group corresponds to one DNA axis (Table 1); within each group, different modality experts (Table 2) contribute differently, and the CGC gate learns the optimal mixture per task:

The four task groups — Engagement, Lifecycle, Value, Consumption — and their constituent tasks are defined in Table 1.

adaTT enforces differentiated transfer: strong intra-group transfer (same DNA perspective) and weaker inter-group transfer (different perspectives, minimizing negative transfer).

Loss-level transfer. A notable departure from the original adaTT [20], which transfers at the representation level, our implementation operates at the *loss level*:

$$\mathcal{L}_i^{\text{adaTT}} = \mathcal{L}_i + \lambda \sum_{j \neq i} w_{i \to j} \cdot \mathcal{L}_j$$

where $w_{i \to j}$ is the transfer weight from task j to task i, computed via gradient cosine similarity between task loss gradients. The base task losses $\mathcal{L}_i$ are weighted by learned uncertainty [18], and binary classification tasks with severe class imbalance (e.g., churn_signal) use focal loss [21] with task-specific α and γ parameters.

This design choice was motivated by two considerations: (1) representation-level transfer requires matching hidden dimensions across heterogeneous experts, which is architecturally cumbersome when experts produce outputs of different shapes and semantics; (2) loss-level transfer naturally respects the task group structure — if two tasks have similar gradient directions (high cosine similarity), their losses reinforce each other, regardless of the expert that produced the prediction.

Note: While theoretically motivated, our ablation (Section 5.3) shows that adaTT **degrades** performance at the 13-task scale due to 156-pair affinity estimation instability. GradSurgery (Section 5.3) was tested as a gradient-level alternative but not adopted due to lack of improvement and VRAM overhead.

Source → Target	Method	Customer Experience	Causal Direction
engagement → nba_primary	hidden_concat	Activity level → purchase probability	Leading indicator
churn → nba_primary	output_concat	Retention risk → acquisition opportunity (class 0 = no NBA)	Inverse correlation

Table 6: Logit transfer relationships reflecting natural customer experience flow.

3.7 Logit Transfer

While adaTT handles symmetric inter-task relationships, logit transfer captures *directional* dependencies that reflect the natural sequence of customer experience:

These transfers connect DNA groups (Table 1): engagement→consumption (intra-Engagement/Consumption axis), lifecycle→consumption (churn→nba_primary, inter-group), reflecting the natural sequence of customer experience. The transfer directions are not learned from data but specified based on domain knowledge of the customer journey. This is a deliberate design choice: while the *strength* of transfer is learned (via adaTT affinity), the *direction* is fixed by business logic, providing an additional layer of interpretability.

3.8 Inherent Explainability

A central claim of this work is that the heterogeneous expert structure provides *structural interpretability at the analytical-perspective level* — gate weights decompose each prediction into named analytical viewpoints (temporal trends, product hierarchy, causal pathways, etc.) as a natural byproduct of the forward pass. This is not causal explanation in the Pearl sense: most experts operate at Level 1 (association) of the causal ladder. The exception is the Causal expert, which learns a NOTEARS DAG and thus approaches Level 2 (intervention) reasoning. When the Causal expert's gate weight is high for a prediction, it indicates not merely correlation but that *structural causal relationships* contributed to the recommendation — a qualitatively stronger form of explanation.

Mechanism. The CGC (Customized Gate Control) module computes an attention over the K experts for each task t. With softmax gating: $w_t = \mathrm{softmax}(W_t \cdot h + b_t) \in \mathbb{R}^K$; with sigmoid gating: $w_{t,k} = \sigma(\boldsymbol{w}_{t,k} \cdot h) / \sum_j \sigma(\boldsymbol{w}_{t,j} \cdot h)$. The weight $w_{t,k}$ directly indicates how much expert k contributed to task t's prediction.

Business interpretability. Because each expert k has a named inductive bias (e.g., k = "Temporal" means time-series patterns, k = "HGCN" means product hierarchy), $w_{t,k}$ carries business meaning without additional interpretation:

Example: For customer c, the recommendation of a commuter check card is driven by:
Temporal (0.35) — card spending has been increasing over 3 months
HGCN (0.28) — merchant category hierarchy shows concentration in transit and convenience MCCs
DeepFM (0.22) — commute-related product × channel interaction pattern
Others (0.15)

What gate weights do and do not explain. Gate weights answer "which analytical perspective mattered most" — not "why this customer will buy this product" in a causal sense. For most experts (Temporal, HGCN, LightGCN, DeepFM, PersLay, OT), the weight indicates associative relevance. For the Causal expert specifically, a high weight signals that a learned DAG structure contributed to the prediction, providing a stronger explanatory basis. This is an honest middle ground: richer than SHAP's feature-level attribution, but short of full counterfactual reasoning.

Comparison with post-hoc methods. SHAP and LIME operate at the *feature* level ("feature_237 contributed 0.12"), requiring an additional mapping step to translate feature attributions into business narratives. Moreover, LIME's local linear approximation is demonstrably unstable — small input perturbations can dramatically change the explanation [34]. In contrast, gate weights are deterministic functions of the input and change smoothly with input variation.

Dual role of features. An important design insight is that features serve two distinct purposes: (1) predictive contribution to AUC, and (2) explanation material for recommendation reasons. Even features with marginal predictive contribution (e.g., TDA topological features may add only $\Delta\mathrm{AUC} \approx 0.01$) provide irreplaceable context for recommendation reasoning. Internally, TDA persistence signals that a customer's spending pattern has a stable topological shape — this is then reverse-mapped via the `interpretation_registry` to business language (e.g., "You maintain a stable transaction pattern") that the customer actually sees. The scientific feature enriches the recommendation context; the customer receives an interpretable description. This dual role motivates our

multi-disciplinary feature engineering and is validated in the ablation study.

4 Training Pipeline

4.1 Config-Driven Design

The entire pipeline is controlled by two YAML files: `pipeline.yaml` (model, training, deployment settings) and `feature_groups.yaml` (feature generation, expert routing). Adding a new dataset, task, or expert requires only configuration changes.

4.2 Data Processing

Following the principle of *pandas-free* data engineering for scalability, the pipeline uses a tiered backend strategy: cuDF (GPU columnar) for generator fitting/transformation, DuckDB [33] (CPU columnar with disk spill) for SQL-based aggregation and I/O, and PyArrow for zero-copy parquet loading into training tensors. pandas is used only as a last-resort fallback for datasets under 10K rows.

This backend choice has practical implications: DuckDB handles the 1M-row dataset with a 4GB memory limit and automatically spills to disk, enabling processing on commodity hardware. cuDF accelerates generator fitting (GMM, HMM, Mamba) on GPU, reducing Phase 0 feature engineering from 30 minutes (pandas) to 8 minutes.

4.3 Data Leakage Prevention

Data leakage is a pervasive risk in multi-stage pipelines where features and labels are processed in separate stages. We implement three structural safeguards:

1. **Generator label exclusion**: Feature generators (GMM, model-derived, etc.) automatically exclude all label columns from their input. Without this guard, we observed XGBoost AUC of 1.0 on the generated features — the generator had encoded label information directly into features.
2. **Temporal split with gap**: A configurable `gap_days` parameter ensures no temporal overlap between training and validation windows. For cross-sectional (single-snapshot) data, the system auto-detects and falls back to random splitting.
3. **Scaler train-only fitting**: StandardScaler is fit exclusively on the training split; validation and test data are transformed using training statistics.
4. **LeakageValidator**: A pre-training check computes Pearson correlation between every feature and every label (on a 50K subsample for efficiency). Correlations above 0.95 trigger a warning and investigation.

5 Experiments

5.1 Benchmark Data

We construct a 1M-customer synthetic benchmark using a four-layer generative model, inspired by the Synthetic Data Vault framework [30] and conditional generative approaches [40], but with a novel *variance budget* mechanism for controllable difficulty:

1. **Latent Personas**: 6 GMM-fitted personas with 5D continuous latent vector (70% persona-conditioned, 30% independent noise).
2. **Gaussian Copula Demographics**: Correlated demographic variables preserving realistic joint distributions [30].
3. **Vectorized Transactions**: Per-customer transaction sequences as LIST columns.
4. **Variance Budget Labels**: Each label's predictability is controlled via $\text{logit} = \sqrt{f_{\text{obs}}} \cdot z_{\text{obs}} + \sqrt{f_{\text{lat}}} \cdot z_{\text{lat}} + \sqrt{f_{\text{noise}}} \cdot \varepsilon$, with post-hoc label noise flipping.

The benchmark underwent several iterations. v2 used uniform-random MCC codes and fixed transaction amounts across personas, producing near-random labels for MCC-dependent tasks. v3 introduced persona-weighted MCC distributions (4–5× boost) with temporal stickiness (30%), persona-dependent transaction amounts, and quantile-based spend_level boundaries, improving MCC task signal but still yielding near-uniform label distributions for acquisition tasks. v4 sharpened persona MCC preferences (8–12× boost),

Tier	**Labels**	f_{obs}	f_{noise}	**XGB AUC**
Easy	segment[2]	determ.	n/a	0.95–1.0
Core	churn_signal	0.04	0.68	0.58–0.65
Hard	will_acquire_*	0.03	0.72	0.50–0.56
V.Hard	next_mcc, top_mcc_shift	0.02	0.78	0.50–0.51

Table 7: Variance budget per label tier. XGB AUC ceiling validates difficulty control.

[2]segment_prediction uses a 4-class customer lifecycle label derived from behavioral clustering, not a deterministic feature transformation. The high XGBoost ceiling reflects strong feature-label correlation but the label requires non-trivial multi-feature interaction to predict, unlike the removed deterministic labels (income_tier, tenure_stage).

increased temporal stickiness to 60%, raised per-product acquisition rates (8–12%), and widened the top_mcc_shift detection window to 30 transactions. v12 (reported here) introduces Financial DNA axis-aligned situation variables — each customer receives an independent situation per DNA axis (engagement: steady/surging/declining/volatile; lifecycle: stable/growing/consolidating/transitioning; value: stable/ascending/descending/shock; consumption: consistent/exploring/focusing/switching). Situations modulate transaction sequence patterns (temporal spacing, amount trends, MCC distribution shifts, product acquisition rates) without altering static demographic features, creating signals recoverable only through Phase 0 generators (Mamba, TDA, HGCN, GMM). Label interactions use 3rd–5th order continuous multiplicative terms with obs_frac=0.15, producing meaningful class distributions for all 13 tasks.

Label signals combine linear observable features (obs_frac=0.15) with latent components (lat_frac=0.35) recoverable through behavioral sequence patterns. Observable interactions are 3rd to 5th order continuous multiplicative (e.g., income × spend × tenure × products × stability for investment propensity), requiring deep nonlinear capacity that tree-based models and shallow networks cannot efficiently approximate. The remaining variance (noise_frac=0.50) is irreducible noise, setting a hard upper bound on model performance.

5.2 Experimental Setup

The ablation validates whether each expert provides a distinct "why" for different task types — confirming that heterogeneous experts are not merely a performance trick but a structural requirement for multi-faceted persuasion.

- **Data**: 1M customers, 349 features (Phase 0 from benchmark_v12), 13 tasks.
- **Hardware**: NVIDIA RTX 4070 (12GB VRAM, 64GB RAM) local.
- **Training**: 10 epochs, batch 5632, lr 0.0005, AMP (FP16), warmup 3 epochs (cosine annealing), no early stopping. adaTT scenarios use warmup=3 epochs, grad_interval=10. GradSurgery scenarios use warmup=2 epochs, conflict_threshold=0.0.
- **Loss weighting**: Uncertainty weighting (Kendall et al.) with per-task loss weights applied on top of learned precision — matching the on-premise reference formula.
- **Metrics**: Metrics are chosen to match each task's production semantic. Binary classification uses AUC (threshold-independent and imbalance-robust). Classification multiclass (segment_prediction, 4 classes) uses F1-macro. Recommendation multiclass (nba_primary with 7 product groups, next_mcc with top-50 merchant categories) uses NDCG@K and top-K accuracy [16], reflecting standard recommendation system evaluation practice. Regression uses MAE. Metrics are reported per task type; a global average across all 13 tasks is avoided because metrics have incompatible semantics across types.

5.3 Joint Feature + Expert Ablation (RQ1 + RQ2)

Bottom-up ablation adds one expert (with its matched feature generator) to the DeepFM baseline; top-down ablation removes one expert from the full 7-expert model. Avg F1m covers segment_prediction (F1-macro); nba_primary and next_mcc use NDCG@3/Acc@3 in per-task reporting.

Expert contribution analysis (Table 8) reveals three patterns. All scenarios use PLE softmax gating with adaTT disabled.

Individual expert contributions show limited differentiation on aggregate AUC with synthetic data. The full 7-expert model (AUC 0.6724) improves only +0.0006 over DeepFM alone (0.6718); task-type-specific contributions (e.g., multiclass NDCG, regression MAE) require per-metric analysis and are expected to diverge on production data with genuine behavioral patterns. In the bottom-up ablation, LightGCN provides the largest AUC gain (+0.0015 over DeepFM baseline), while in the top-down ablation, no single expert removal causes AUC to drop below the DeepFM-only level. This confirms that the synthetic benchmark's formula-based feature–label relationships do not require specialized expert representations — a limitation expected to reverse with production data where genuine temporal, hierarchical, and collaborative signals exist.

Causal and OT experts cause segment_prediction negative transfer. Both DeepFM+Causal and DeepFM+OT show F1-macro drops of −0.04 (†marked in table), entirely attributable to segment_prediction F1 collapsing from 0.40 to 0.28. The identical magnitude for two architecturally different experts (NOTEARS DAG vs. Sinkhorn Wasserstein) indicates a *softmax gate redistribution effect*: adding any expert that receives the full feature vector forces the gate to redistribute weight away from DeepFM, regardless of the new expert's architecture.

HGCN removal degrades nba_primary ranking. While aggregate AUC is unaffected, per-task analysis shows that removing HGCN causes nba_primary NDCG@3 to drop by −0.015 — the largest per-task degradation in the top-down ablation, confirming that

Scenario	Avg AUC	Avg F1m	Avg MAE	Val Loss
Baselines				
Full (7 experts, PLE softmax)	0.6724	0.2010	0.9596	25.72
DeepFM only	0.6718	**0.2031**	0.9591	25.74
Bottom-up: DeepFM + single expert (sorted by AUC desc)				
DeepFM + LightGCN	**0.6733**	0.2000	0.9598	25.75
DeepFM + HGCN	0.6725	0.1980	**0.9577**	25.74
DeepFM + TDA	0.6723	0.2025	0.9603	25.72
DeepFM + Temporal	0.6720	0.2015	0.9593	25.70
DeepFM + Causal	0.6715	0.1609†	0.9587	25.74
DeepFM + OT	0.6714	0.1596†	0.9591	25.75
Top-down: Full minus one expert				
Full − LightGCN	0.6727	0.2016	0.9594	25.74
Full − TDA	0.6726	0.2012	0.9582	25.73
Full − Temporal	0.6725	0.2012	0.9596	25.71
Full − HGCN	0.6724	0.2014	**0.9586**	25.72
Full − Causal	0.6724	0.2010	0.9617	25.73
Full − OT	0.6721	0.2020	0.9591	23.60

Table 8: Joint feature + expert ablation results. Bold = best in group. †Anomalous F1-macro drop from segment_prediction negative transfer.

product hierarchy encoding is essential for recommendation ranking.

5.4 Task × Structure Cross Ablation (RQ3)

Nine structure variants are compared across three dimensions: gate type (shared-bottom / softmax / sigmoid), inter-task transfer (none / adaTT loss-level / GradSurgery gradient-level), and uncertainty weighting (all variants use the corrected implementation where per-task loss weight is applied on top of learned precision). All variants use the full 7 heterogeneous expert basket with 10 epochs, batch size 5632, and AMP enabled. Avg AUC covers binary tasks only; NDCG@3 is reported for nba_primary (7-class product recommendation); Avg F1m covers multiclass tasks; Avg MAE covers regression tasks.

Three findings emerge from the structure ablation.

Finding 1: PLE softmax outperforms sigmoid in heterogeneous MTL. This reverses the conventional wisdom from homogeneous-task PLE literature [39],

Variant	Avg AUC	Best NDCG@3	Final NDCG@3	Avg F1m	Avg MAE
Gate type comparison (no inter-task transfer)					
Shared Bottom	0.6711	0.7014	0.6831	0.2007	0.9602
PLE Softmax	**0.6729**	**0.7144**	0.6814	0.2009	**0.9598**
PLE Sigmoid	0.6728	0.7131	0.6820	**0.2021**	0.9601
adaTT loss-level transfer					
SB + adaTT	0.6698	0.7048	0.6921	0.1963	0.9672
Sigmoid + adaTT	0.6541	0.6998	0.6762	0.1992	0.9721
Softmax + adaTT	0.6549	0.7022	0.6780	0.1993	0.9709
GradSurgery gradient-level projection†					
SB + GradSurgery (exp.)	0.6704	0.6963	0.6860	0.1986	0.9614
Softmax + GradSurgery (exp.)	0.6726	0.6918	0.6830	**0.2027**	0.9588
Sigmoid + GradSurgery (exp.)	0.6721	0.6888	0.6811	0.2020	0.9603

Table 9: Structure ablation across gate type and inter-task transfer variants. Bold = best per metric within gate comparison. †GradSurgery uses batch 4096 (vs. 5632) due to retain_graph VRAM overhead.

where sigmoid gates are preferred for their non-competitive expert weighting. In our 13-task setting with 7 binary + 3 multiclass + 3 regression tasks, softmax's competitive selection *protects* minority-type tasks (multiclass) by isolating expert assignments, preventing binary-task gradients from corrupting multiclass-relevant experts. Crucially, the uncertainty weights converge to identical values (nba_primary: 0.335) under both gate types, confirming that the improvement is purely structural.

Finding 2: adaTT loss-level transfer degrades all metrics. Adding adaTT causes AUC to drop by −0.019 and best NDCG@3 by −0.013 relative to the corresponding PLE-only variant (sigmoid+adaTT vs. sigmoid alone). The root cause is a *scaling mismatch*: 13 tasks produce 156 directed transfer pairs, but 7 active transfer epochs (10 total minus 3 warmup) provide insufficient gradient samples for stable affinity estimation. The original adaTT was validated on 4 tasks (12 pairs); our 13× increase in pair count without proportional epoch increase causes noisy transfer that acts as gradient noise.

Finding 3: Correcting uncertainty weighting yields larger gains than architecture changes. The single largest performance improvement (+0.018 NDCG@3, +0.031 F1-macro) came not from gate type or transfer mechanism, but from fixing a subtle implementation gap where per-task loss weights were silently ignored when uncertainty weighting was active. This finding underscores that *loss balancing correctness* can dominate *architectural sophistication* in heterogeneous MTL — a practical lesson often overlooked in architecture-focused papers.

5.5 Expert Contribution Analysis: Beneficial vs. Negative Transfer (RQ4)

We assess each expert's contribution by examining how performance changes when it is individually removed from the full model (baseline AUC = 0.6724, joint_full 10-epoch). Positive ΔAUC indicates the expert contributes negative transfer; negative ΔAUC indicates the expert is beneficial. Magnitudes are small (< 0.001) due to synthetic data limitations; genuine expert differentiation requires production data with real behavioral signals.

Removed Expert	ΔAUC	Interpretation
−LightGCN	+0.0003	marginal negative transfer
−TDA	+0.0002	marginal negative transfer
−Temporal	+0.0001	negligible
−Causal	+0.0001	negligible
−HGCN	0.0000	neutral
−OT	**−0.0004**	marginal benefit

Table 10: Expert contribution analysis: ΔAUC relative to full 7-expert model (AUC = 0.6724). Values from top-down ablation.

Expert ΔAUC magnitudes are uniformly small (< 0.001), reflecting the synthetic benchmark's limited capacity to differentiate experts. No expert is clearly beneficial or harmful at this scale. This near-uniform result requires careful interpretation along three dimensions.

Synthetic data limitation. The negative transfer from Temporal and TDA is attributable to the synthetic benchmark's data characteristics. Synthetic transaction sequences lack genuine temporal patterns — MCC codes are generated from persona-weighted distributions with stickiness, but without real behavioral dynamics such as seasonality, life-event triggers, or spending regime changes. When an expert's inductive bias (temporal dynamics, topological persistence) has no matching signal in the data, it injects noise into the shared representation. In production data where genuine behavioral sequences carry predictive signal, these experts' contributions are expected to change substantially.

Single-metric blind spot. Aggregate AUC averages across 7 binary tasks and does not capture contributions to multiclass ranking (NDCG@3) or regression (MAE). An expert may reduce binary AUC while improving multiclass or regression performance — effects invisible in the aggregate. Per-task-type analysis is required before concluding an expert is dispensable.

Diagnostic value of asymmetry. The most important observation is not *which* experts are harmful, but that the heterogeneous design *enables this diagnosis*. A homogeneous MLP expert pool would show uniform, undifferentiated degradation — every expert would degrade similarly because they share the same inductive bias. The heterogeneous basket produces an *actionable diagnostic*: "Temporal is harmful in synthetic data; HGCN is structurally essential" is a statement that directly informs architecture decisions. This diagnostic capability is itself a contribution of the heterogeneous expert design,

Task	Layer 1 $H/\log K$	Layer 2 $H/\log K$	Interpretation
top_mcc_shift	0.347	0.542	Single expert dominance
product_stability	0.428	0.705	Concentrated routing
segment_prediction	0.612	0.332	Layer 2 concentrates
nba_primary	0.877	0.724	Diverse expert usage
will_acquire_payments	0.882	0.688	Diverse expert usage
CGC Attention (all tasks)	1.000	1.000	Uniform (undifferentiated)

Table 11: Per-task CGC gate entropy ratios ($H_t/\log K$; higher = more uniform).

independent of whether any individual expert improves aggregate AUC.

5.6 Explainability Analysis (RQ5)

The CGC gate produces interpretable routing weights that enable direct attribution of "which expert contributed how much" per task. Under the recommended softmax configuration, weights sum to one, making relative contributions directly comparable. We examine per-task gate weight distributions across all 13 tasks to identify (a) which experts dominate which task types, and (b) whether the learned routing aligns with domain intuition (e.g., temporal expert weighted highly for churn prediction, causal expert for intervention-sensitive tasks).

Gate weights from the CGC extraction layers provide per-task expert utilization profiles. Tasks with low entropy ratio (e.g., top_mcc_shift at 0.347) concentrate on 1–2 experts, providing clear attribution: the recommendation is driven primarily by the dominant expert's feature group. Tasks with high entropy (e.g., will_acquire_payments at 0.882) leverage diverse experts, requiring multi-factor explanation. This entropy-based explainability directly maps to the Financial DNA task groups and enables the rule-based fallback (Layer 3) to select appropriate explanation templates per task.

Detailed per-task gate weight profiles and dominant expert assignments are provided in the Gate Entropy Analysis (RQ6) below, which jointly addresses both routing collapse and task-expert specialization.

5.7 Gate Entropy Analysis (RQ6: Does routing collapse occur?)

Beyond function space collapse (prevented structurally by heterogeneous experts), routing collapse can occur when CGC gate weights concentrate on a single expert. We measure per-task gate entropy:

$$H_t = -\sum_k w_{t,k} \log w_{t,k}$$

where $w_{t,k}$ is the gate weight for expert k on task t. Maximum entropy $\log K$ (K=7 experts, $\log 7 \approx 1.95$) indicates uniform expert utilization; entropy near zero indicates routing collapse. We report normalized entropy ratio $\frac{H_t}{\log} K \in [0, 1]$ to facilitate comparison across different numbers of experts.

5.7.1 CGC Layer Gate Specialization

Measured on the PLE-softmax checkpoint, CGC layer gate weights show meaningful task-level specialization with entropy ratios ranging from 0.33 to 0.88 across the 13 tasks and 2 CGC layers. This confirms that routing collapse has not occurred: all tasks engage multiple experts, while distinct tasks show distinct routing patterns. Low-entropy tasks concentrate on 1–2 experts; high-entropy tasks draw broadly across the expert basket. CGC attention weights remain perfectly uniform across all tasks, indicating that expert selection operates at the extraction layer, not at the attention aggregation layer.

The divergence between extraction-layer and attention-layer routing is structurally informative. *Low-entropy tasks* (top_mcc_shift: 0.347, product_stability: 0.428, segment_prediction layer 2: 0.332) concentrate predominantly on 1–2 experts, suggesting that these tasks are solved by a narrow analytical lens (e.g., temporal trend or graph structure alone). *High-entropy tasks* (will_acquire_payments: 0.882, nba_primary: 0.877) draw broadly from the expert basket, consistent with the nature of next-best-action and payment acquisition tasks that integrate multiple behavioral signals simultaneously.

Notably, CGC *attention* weights remain perfectly uniform (entropy ratio = 1.000 across all tasks and both layers), indicating that the attention aggregation mechanism has not differentiated — functionally equivalent to simple averaging of expert outputs. This suggests the model relies on the extraction-layer gates to perform meaningful routing, while the attention mechanism acts as a residual blending layer. Future work may benefit from regularizing or removing the attention aggregation to reduce parameter redundancy.

6 Discussion

6.1 Findings Summary

The ablation experiments yield the following principal findings.

Loss balancing correctness dominates architecture choice. The single largest performance improvement across all experiments came from correcting a subtle implementation gap in uncertainty weighting (Kendall et al.), where per-task loss weights from the pipeline configuration were silently ignored. The corrected formula — loss_weight × (precision × L + log_var) with log_var clamped to $[-4, 4]$ and precision clamped to $[10^{-3}, 100]$ — matches the on-premise reference implementation and yielded +0.018 NDCG@3 and +0.031 F1-macro on the shared-bottom baseline alone. This underscores that in heterogeneous MTL with mixed loss types, *getting the loss weighting right* matters more than architectural sophistication.

Softmax gating outperforms sigmoid in heterogeneous MTL. This finding contradicts the conventional preference for sigmoid gating in PLE literature [39] and recent sigmoid MoE work [28]. In our 13-task setting (7 binary + 3 multiclass + 3 regression), softmax's competitive expert selection *protects* multiclass tasks by assigning dedicated experts, while sigmoid's cooperative blending allows binary-task gradients to corrupt multiclass-relevant representations. The effect is metric-specific: softmax leads in NDCG@3 (+0.013 over shared-bottom) while sigmoid and softmax are near-identical in AUC. We attribute the reversal to *task-type heterogeneity*: when all tasks share the same loss type (as in prior work), cooperative blending is beneficial; when loss types differ fundamentally, competitive isolation is protective.

Loss-level transfer (adaTT) fails at 13-task scale. adaTT degrades all metrics relative to the corresponding PLE-only variant. The mechanism — adding weighted auxiliary losses $L_i^{\text{adaTT}} = L_i + \lambda \sum_{j \neq i} w_{i \to j} L_j$ — amplifies gradient noise when affinity estimation is unstable. With 156 directed task pairs and only 7 active transfer epochs (10 total minus 3 warmup), the per-pair gradient cosine similarity measurements are insufficiently averaged. This identifies a *scalability boundary* for loss-level transfer: adaTT was validated on 4 tasks (12 pairs) [20]; scaling to 13 tasks (156 pairs) without proportional epoch increase exceeds the method's estimation capacity.

GradSurgery: gradient-level projection as an alternative. To address the loss-level transfer limitation, we tested task-type gradient surgery (GradSurgery), which replaces loss-level transfer with gradient-level conflict resolution. Instead of estimating 156 pair-wise affinities, GradSurgery aggregates task gradients into 3 type-groups (binary, multiclass, regression) and applies PCGrad-style projection only between conflicting groups. This reduces the estimation problem from $O(k^2)$ to $O(3^2) = O(1)$ relative to the number of tasks. Softmax+GradSurgery achieves AUC 0.6726 (vs. softmax-alone 0.6729) and the highest F1-macro (0.2027) and lowest MAE (0.9588) across all variants, while best NDCG@3 (0.6918) is lower than softmax-alone (0.7144). Critically, GradSurgery does *not* degrade performance — unlike adaTT, which drops AUC by −0.019. However, the retained computation graph overhead constrains batch size on 12GB GPUs, the NDCG gap suggests that gradient-level projection may not fully capture the ranking signal that PLE's representation-level gate routing provides, and GradSurgery shows no meaningful advantage over PLE softmax alone. GradSurgery was therefore not adopted for production. The most robust configuration remains PLE softmax without transfer mechanisms.

Operational motivation validated. The shared-bottom baseline (AUC 0.6711, NDCG@3 0.7014) already exceeds per-task XGBoost ceilings on all 13 tasks, validating the operational motivation for MTL consolidation: a single model replaces 13 individual models with no performance penalty, enabling unified training, serving, and monitoring. PLE softmax provides incremental gains (NDCG@3 +0.013) at no additional serving cost.

6.2 Training Dynamics: Loss–Metric Decoupling and Epoch Selection

A 30-epoch training run (cosine LR with $T_0 = 10$, warm restarts) reveals a systematic decoupling between validation loss and discriminative metrics that has practical implications for best-model selection in heterogeneous MTL.

Note: This 30-epoch diagnostic run is separate from the 10-epoch ablation results reported in Tables 8–9. It was conducted to analyze convergence dynamics and does not affect the reported ablation numbers.

Observation: metrics peak at epoch 10, not epoch 30. Validation loss decreases monotonically across all 30 epochs (32.11 → 22.68, −29.4%), suggesting continued learning by a standard stopping criterion. However, discriminative metrics peak at epoch 10 and subsequently degrade: AUC peaks at epoch 10 (0.6726) and falls to 0.6687 at epoch 30 (−0.4pp), while NDCG@3 peaks at epoch 10 (0.6976) and drops to 0.6673 at epoch 30 (−3.0pp).

Mechanism: regression task dominance of the loss function. The aggregate validation loss is dominated by the three regression tasks, which produce large

MAE values relative to binary cross-entropy magnitudes. Continued training genuinely improves regression MAE ($0.9596 \rightarrow 0.9580$ at epoch 30), but at the cost of overfitting to synthetic benchmark noise in the classification and ranking tasks. The benchmark's formula-based feature–label relationships are fully learned by epoch 10; subsequent epochs overfit to residual noise that has no counterpart in the discriminative metrics.

Cosine LR restarts amplify the effect. The cosine schedule with $T_0 = 10$ causes sharp metric oscillations at cycle boundaries (epochs 11, 20, 29): NDCG@3 drops abruptly at each restart as the learning rate spikes and disrupts the representation alignment that the gate weights have accumulated. This suggests that cosine warm restarts, while beneficial for escaping loss plateaus, may periodically damage ranking-relevant representations in the heterogeneous setting.

Implication: composite metric for best-model selection. Validation loss alone is an unreliable stopping criterion when task types are heterogeneous and regression loss dominates. We recommend selecting the best checkpoint using a composite metric — e.g., $\alpha \cdot \text{AUC} + \beta \cdot \text{NDCG} + \gamma \cdot (1 - \text{MAE})$ — rather than minimum val_loss. In our setting, this corresponds to selecting epoch 10 over epoch 30, recovering +0.4pp AUC and +3.0pp NDCG@3.

6.3 Practical Implications

Resource-constrained development. This system was built without dedicated ML infrastructure budget, on a single desktop GPU, by a three-person team augmented with AI development agents. This demonstrates that complex multi-task recommendation systems are no longer exclusive to organizations with large ML teams and GPU clusters. The key enablers were: (1) config-driven architecture minimizing code changes, (2) AI agents handling parallel implementation tasks under human architectural guidance, (3) heterogeneous expert design achieving expressiveness through structural bias rather than parameter scale, and (4) knowledge distillation eliminating GPU serving costs.

Defense over offense in production ML. In financial recommendation services, maintaining AUC above an operational threshold matters far more than pushing it higher. The heterogeneous expert design directly supports this *defensive* posture: expert contribution diagnostics (ablation reveals which inductive biases are beneficial vs.
harmful per data regime), drift detection triggering automatic retraining, and the champion-challenger gate requiring manual approval before deployment. The architecture prioritizes *not getting worse* over *getting better* — a perspective that aligns with financial regulators' emphasis on model risk management and operational resilience.

Feature engineering philosophy. A deeper lesson from this work is that *what to observe* matters more than *how to model*. The heterogeneous expert architecture is effective not because of architectural novelty alone, but because each expert is paired with features that ask a **specific question** about the customer: "Is their spending decaying with a half-life?" (chemical kinetics), "Is product adoption spreading like contagion?" (SIR epidemiology), "Is their income permanent or transitory?" (Friedman PIH [10]). The model merely combines answers to these questions. If the questions are limited to conventional statistics — mean, variance, trend — even the most sophisticated architecture cannot capture the structural complexity of customer behavior. This suggests that multi-disciplinary feature engineering, often treated as preprocessing, deserves elevation to a first-class architectural design decision, co-designed with the expert types that will consume each feature group.

Generalizability beyond finance. While this work targets financial product recommendation, the underlying principle — understanding a single user from multiple perspectives simultaneously via heterogeneous experts — applies to any domain where personalization requires multi-faceted user modeling. Healthcare (diagnosis + treatment + risk + lifestyle), education (knowledge level + learning style + engagement + career path), and insurance (risk assessment + product fit + claim prediction + retention) all face structurally similar challenges: multiple interdependent prediction tasks about the same individual, where different analytical lenses capture different aspects. We expect the heterogeneous expert PLE pattern to generalize to these domains, with domain-specific expert types replacing the financial experts while retaining the structural benefits of collapse resistance and inherent explainability.

Lessons from design iteration. The final architecture emerged through repeated trial and rejection. The first candidate — Black-Litterman Bayesian model combination — was abandoned because the blended posterior made individual model contributions opaque, failing the explainability requirement that drives the entire project. The second candidate — N-model ensemble — was rejected for N× management overhead. These failures led to the key reframing: combine experts *inside* a single model, not *outside* it.

An early ablation (before the uncertainty weighting fix) showed sigmoid outperforming softmax, which appeared to confirm the NeurIPS 2024 finding [28] that

softmax competition harms heterogeneous experts. However, after correcting the loss weighting implementation, softmax reversed to outperform sigmoid on the primary ranking metric (NDCG@3), demonstrating that **loss balancing bugs can masquerade as architectural findings** — a cautionary lesson for MTL practitioners.

A particularly instructive failure: a configuration bug that disabled PLE collapsed the 7-expert basket into a single MLP, making all 24 ablation scenarios produce identical AUC (0.913).[3] This was only discovered through systematic result comparison — reinforcing the principle that ablation results must be verified against expected variation before drawing conclusions.

Infrastructure choice: extending GPU ROI beyond LLM serving. Financial institutions have increasingly invested in GPU infrastructure, but predominantly for generative AI workloads — chatbots, document summarization, and LLM-based customer service. Recommendation model training remains largely CPU-based (XGBoost, LightGBM), leaving GPU capacity underutilized outside of LLM inference.

Our architecture demonstrates that the same GPU hardware already deployed for LLM serving can simultaneously support 13-task MTL training, requiring only 12GB VRAM on a single consumer-grade GPU (RTX 4070). This extends the return on existing GPU investments without additional procurement — a practical consideration for institutions where new hardware acquisition requires multi-month approval cycles. For institutions without on-premise GPU, our SageMaker + Lambda serverless architecture provides an alternative: GPU Spot instances (60–70% discount) are provisioned on-demand for training and released immediately after, while serving runs on CPU-only Lambda functions via knowledge distillation to LGBM (detailed in companion paper).

Operational simplicity. The config-driven design (two YAML files control the entire pipeline) enables teams with 1–2 ML engineers to operate the system. Adding a new product category, task, or feature group requires only configuration changes, not code modifications. This stands in contrast to model ensemble approaches where each model is a separate management point with its own training pipeline, serving endpoint, and monitoring dashboard.

6.4 Limitations

Synthetic benchmark only. All experiments use a 1M-customer synthetic benchmark with formula-based feature–label relationships. While the benchmark controls difficulty via a variance budget mechanism and validates against XGBoost AUC ceilings, it cannot reproduce the complex cross-feature interactions present in real financial customer data. In particular, the formula-based label generation may understate the benefit of expert specialization — real data where temporal patterns, graph structures, and causal pathways carry genuinely different signals may amplify PLE's advantage over shared-bottom. Validation on proprietary financial data is planned for a subsequent version but is not included here due to regulatory data access constraints.

adaTT evaluation at limited epoch budget. adaTT's degradation may be partly attributable to the 10-epoch training budget. With 50+ epochs, affinity estimation may stabilize sufficiently for 156 task pairs. However, the 10-epoch budget reflects practical constraints (ablation throughput on a single GPU) and is sufficient for convergence of non-transfer variants.

Single-GPU training. The current implementation runs on a single GPU (RTX 4070, 12GB). DDP support is architecturally designed but not yet experimentally validated.

LLM dependency. Recommendation reason generation relies on LLM inference, introducing latency and cost trade-offs (detailed in companion paper).

6.5 Ethics and Data Statement

All experiments in this version use **fully synthetic benchmark data** generated via Gaussian Copula with a fixed random seed (seed=42). No real customer data is included in this version. Validation on production data is planned for a subsequent revision. The benchmark data generator and all configuration files are publicly available in the repository. The system is designed for deployment on low-risk check card products only; investment and insurance recommendations are explicitly excluded from the operational scope (see companion paper, Section 6.3).

6.6 Future Work: Scaling Considerations

The current architecture is intentionally lightweight — each expert uses a compact, domain-specific design (e.g., 2-layer MLP for DeepFM deep branch, pre-computed embeddings for HGCN) rather than scaling parameters. This is a deliberate design choice, not merely a resource constraint: structural inductive biases substitute for raw parameter count.

[3] The 0.913 AUC reflects an earlier benchmark version (v4) with different data distributions; current benchmark v12 produces AUC in the 0.67 range.

When scaling to larger institutions with dedicated GPU infrastructure, the natural progression is:

- **Data enrichment** — longer transaction histories, broader product coverage, richer interaction signals — before model capacity increase;
- **Per-expert input dimension expansion** — wider feature groups feeding each expert's specialized inductive bias;
- **Expert depth/width scaling** — deeper layers within individual experts, particularly Temporal (longer sequence modeling) and HGCN (deeper hierarchy encoding);
- **Multi-GPU training** via DDP, which is architecturally supported but not yet validated.

Notably, scaling the expert basket size (adding more expert types) is less promising than scaling individual expert capacity, because the heterogeneous design already covers the major mathematical perspectives relevant to financial behavior.

7 Conclusion

We presented Heterogeneous Expert PLE, a multi-task learning architecture for financial product recommendation that consolidates 13 heterogeneous tasks (7 binary + 3 multiclass + 3 regression) into a single model, replacing 13 individual models while maintaining or exceeding per-task performance.

Three contributions emerge from this work. **First**, the heterogeneous expert basket — seven architecturally distinct experts with FeatureRouter-based input specialization — provides structural collapse resistance and inherent explainability through business-interpretable gate weights, without reliance on post-hoc attribution methods.

Second, ablation at 13-task scale reveals that loss-level inter-task transfer (adaTT) degrades performance due to affinity estimation instability at 156 task pairs, while the single largest improvement comes from *correcting loss balancing* rather than architectural changes. This finding challenges the common assumption that architectural sophistication is the primary lever for MTL improvement, and highlights that implementation correctness of loss weighting deserves the same scrutiny as architecture design.

Third, we tested GradSurgery, a task-type gradient projection method that addresses inter-task interference at the gradient level rather than the loss level, reducing the $O(k^2)$ task-pair estimation problem to $O(1)$ type-group projections. GradSurgery avoids the degradation caused by loss-level transfer (F1-macro 0.203 vs. adaTT's 0.199; MAE 0.959 vs. 0.971), but does not surpass PLE softmax alone on the primary ranking metric (NDCG@3 0.692 vs. 0.714) and incurs significant VRAM overhead from the retained computation graph. GradSurgery is therefore not adopted for production. The most effective configuration is therefore PLE softmax *without* transfer mechanisms — structural expert isolation through gating proves more robust than numerical gradient manipulation.

Softmax gating outperforms sigmoid in this heterogeneous setting, reversing findings from homogeneous-task literature — a result we attribute to softmax's protective isolation of minority-type tasks (multiclass) from majority-type gradients (binary).

The architecture, benchmark data, and ablation framework are released as open source.[4] A companion paper addresses the downstream pipeline: knowledge distillation [15] to LGBM [17], multi-agent recommendation reason generation, and regulatory compliance mapping for Korean FSS and EU AI Act requirements.

Author Contributions

Seonkyu Jeong (PM / Lead Architect / Data Scientist): Conceived the project direction, designed the heterogeneous expert architecture and two-axis decomposition framework, selected multi-disciplinary feature engineering approaches based on structural isomorphism, defined task groups and regulatory compliance mapping, led AI-augmented development methodology, and wrote the manuscript. Overall technical leadership and Scrum-based rapid feedback coordination.

Euncheol Sim: Data ingestion pipeline, feature engineering implementation, feature business reverse-mapping, and vector database pipeline management.

Youngchan Kim: Model training, mathematical verification, and knowledge distillation implementation.

All authors collaborated through Scrum sprints with rapid feedback cycles.

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[4] https://github.com/bluethestyle/aws_ple_for_financial

opment was conducted on a single desktop-grade GPU (NVIDIA RTX 4070, 12GB VRAM) in a repurposed, inadequately ventilated workspace with no dedicated ML infrastructure budget, no institutional network support, and no institutional cloud computing allocation. Data collection was constrained to a legacy HIVE environment with no access to Spark or Impala, requiring custom parallel query logic to overcome I/O bottlenecks.

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Appendix

A. Configuration Schema

The entire system is controlled by two YAML configuration files: `pipeline.yaml` (model architecture, training hyperparameters, task definitions, AWS deployment settings) and `feature_groups.yaml` (feature group definitions, generator parameters, expert routing rules). Full configuration files are available in the accompanying repository.

B. Ablation Scenario Definitions

The 27 ablation scenarios (18 joint feature+expert + 9 structure cross) are organized into two phases:

Phase 1 — Feature + Expert Joint Ablation (18 scenarios): Baseline (DeepFM only with base features), DeepFM with all features, full model, 8 bottom-up scenarios (DeepFM + one expert with matching features), and 6 top-down scenarios (full minus one expert-feature pair).

Phase 2 — Task x Structure Cross Ablation (9 scenarios): Nine structural variants — shared-bottom (no PLE/adaTT), PLE-softmax, PLE-sigmoid, adaTT-only, PLE-softmax+adaTT, PLE-sigmoid+adaTT, and GradSurgery×3 (softmax+GradSurgery, sigmoid+GradSurgery, GradSurgery-only) — all using the full 13 tasks and 7 heterogeneous experts with 10 epochs.

C. Benchmark Data Generation

The synthetic benchmark uses a four-layer generative model: (1) latent personas via 6-component GMM with 5D continuous latent vector (70% persona-conditioned, 30% independent noise); (2) Gaussian Copula demographics preserving realistic correlations; (3) vectorized transaction sequences as LIST columns; (4) variance-budget labels with post-hoc noise flipping (6–8%). Seed=42 ensures reproducibility. Full generation code is available in the accompanying repository.

D. AMP FP16 NaN Diagnosis and GradScaler Tuning

Heterogeneous experts with on-prem-aligned activation functions (ODE-based LNN, linear HGCN output, Softplus TDA weights) produce wider intermediate value ranges than homogeneous MLPs. Under AMP (FP16), this initially caused GradScaler overflow that cascaded into NaN loss — observed from epoch 2 in PLE configurations with default GradScaler settings. All reported experiments use AMP FP16 training, which provides approximately 2× throughput on the T4/RTX GPU. The NaN issues were resolved by tuning three GradScaler parameters: `init_scale=1024` (lower initial scale reduces overflow at startup), `growth_interval=2000` (slower scale growth prevents rapid re-escalation after overflow events), and `max_scale=4096` (hard ceiling prevents runaway scaling for wide-range expert activations). Additionally, per-task log-variance parameters in the uncertainty weighting head are clamped to $[-5, 5]$ to prevent Softplus overflow from propagating through the loss. With these mitigations, zero NaN batches are observed across all ablation runs.

E. Structural Isomorphism Verification

To address the concern that cross-disciplinary feature engineering may be "plausible analogy rather than rigorous isomorphism," we present the governing equations side by side:

Chemical Kinetics -> Spending Activation: Arrhenius: $k = Ae^{-E_a/RT}$ where E_a = activation energy. Spending: $p_{\text{reactivate}} = Ae^{-E_{\text{threshold}}/S}$ where S = stimulus intensity. Both describe the probability of a state transition as a function of an energy barrier and an external driving force.

SIR -> Product Adoption: Epidemiology: $d\frac{I}{d}t = \beta SI - \gamma I$ Adoption: $d\frac{A}{d}t = \beta_{\text{exposure}} \cdot U \cdot A - \gamma_{\text{churn}} \cdot A$ where U = unadopted customers, A = adopted, β = contact/exposure rate. The compartmental dynamics are identical; only the variable names change.

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